

PUERTO PLATA LUPERON, Dominican Republic

WMO: **784570**Lat: **19.758N**Lon: **70.570W**Elev: **15**StdP: **14.69**Time Zone: **-4.00 (W04)**Period: **00-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	66.6	67.8	63.9	88.9	70.8	65.0	92.5	70.6	19.0	81.6	18.4	81.1	1.1	180	0.313

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	93.1	80.6	91.6	80.3	91.0	80.2	82.6	89.6	81.9	89.0	81.3	88.6	14.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	159.8	88.1	80.1	156.1	87.5	79.1	151.0	86.7	46.4	89.7	45.7	89.7	45.1	89.3	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.8	17.8	63.8	97.3	1.6	2.4	62.7	99.0	61.7	100.4	60.8	101.8	59.7	103.5		
(5)			61.7	84.6	2.1	2.2	60.2	86.2	58.9	87.5	57.8	88.7	56.2	90.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	75.8	76.2	76.6	78.4	80.1	82.5	83.0	83.1	82.7	81.8	79.2	77.3
	DBStd	3.30	1.90	1.92	2.39	1.96	1.93	1.75	1.62	1.68	1.54	1.79	1.96	1.89
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10858	800	733	824	851	935	975	1022	1026	982	986	877	847
	CDD65	5383	335	313	359	401	470	525	557	561	532	521	427	382
	CDH74	49223	2055	2041	2554	3211	4229	5668	6216	6307	5791	5195	3409	2547
	CDH80	17465	442	459	646	935	1416	2291	2525	2607	2423	2029	1054	638
Wind	WSAvg	7.1	5.8	6.9	7.3	7.8	7.8	8.4	9.0	8.6	7.0	5.9	5.5	5.7
Precipitation	PrecAvg	69.8	8.6	5.8	5.9	6.1	4.8	1.9	2.3	3.1	2.8	5.3	11.6	11.6
	PrecMax	92.6	22.3	12.4	18.9	14.4	18.0	4.8	7.2	8.4	6.6	14.7	21.8	32.0
	PrecMin	42.4	1.9	0.1	0.3	0.2	0.6	0.2	0.4	0.8	0.6	0.3	2.2	1.7
	PrecStd	13.9	5.0	3.6	4.1	4.0	4.6	1.3	1.9	2.0	1.6	3.3	5.2	7.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.8	87.9	89.4	89.8	91.6	94.6	93.5	93.5	94.7	93.5	90.0	88.0
		MCWB	75.9	76.9	76.4	77.7	78.6	80.5	80.2	80.8	80.7	80.5	79.4	77.7
	2%	DB	85.7	85.9	86.4	87.8	89.5	91.7	91.7	91.7	91.7	91.4	88.1	86.2
		MCWB	76.3	76.5	76.8	78.2	78.9	80.3	80.1	80.6	80.8	80.5	79.5	77.4
	5%	DB	84.1	84.3	85.6	86.2	87.9	90.0	90.9	91.1	90.6	89.8	87.3	84.6
		MCWB	76.1	75.9	76.6	77.5	78.9	80.0	80.0	80.5	80.5	80.2	79.3	76.9
	10%	DB	82.5	82.7	83.9	84.6	86.3	89.4	89.6	89.7	89.6	88.2	85.9	83.9
		MCWB	75.4	75.1	75.8	76.7	78.4	79.9	79.8	80.2	80.3	79.9	78.5	76.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.7	79.2	79.3	80.6	81.6	83.5	82.8	83.2	83.2	82.7	81.5	79.8
		MCDB	84.2	85.0	85.5	86.5	87.2	89.9	89.8	90.4	90.6	89.7	86.5	84.6
	2%	WB	77.5	77.7	78.0	79.2	80.4	82.1	81.8	82.2	82.2	81.7	80.3	78.7
		MCDB	83.2	84.0	84.5	85.5	86.9	89.1	89.2	89.7	89.6	88.8	86.3	84.2
	5%	WB	76.8	76.6	77.0	78.2	79.7	81.0	81.0	81.5	81.5	80.9	79.6	77.8
		MCDB	82.5	82.9	83.8	84.7	86.2	88.4	88.6	89.0	88.9	88.0	85.7	83.5
	10%	WB	75.9	75.7	76.1	77.3	78.8	80.2	80.2	80.7	80.8	80.1	78.8	77.0
		MCDB	81.6	81.7	82.6	83.9	85.4	87.6	87.7	88.2	88.2	87.1	85.0	82.8
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.1	14.0	13.6	13.7	14.6	14.2	14.2	15.0	14.3	13.1	13.2
		MCDBR	16.3	16.1	16.4	15.3	15.2	15.8	15.6	15.7	16.5	15.9	15.1	14.9
		MCWBR	8.8	8.5	8.3	7.7	7.6	7.6	6.8	6.8	7.7	7.8	7.8	7.9
	5% WB	MCDBR	15.0	15.3	15.2	14.4	14.4	15.2	14.5	14.5	15.6	15.0	14.3	14.0
		MCWBR	9.0	8.8	8.5	7.9	8.0	8.1	7.4	7.3	8.1	7.9	8.1	8.2
Clear-Sky Solar Irradiance	taub	0.371	0.386	0.399	0.428	0.458	0.524	0.510	0.484	0.436	0.397	0.390	0.392	
	taud	2.552	2.494	2.454	2.397	2.333	2.099	2.141	2.240	2.412	2.528	2.538	2.503	
	Ebn at Noon	288	289	289	279	268	248	252	260	273	282	280	277	
	Edh at Noon	30	34	36	39	41	51	49	45	37	32	31	31	
All-Sky Solar Radiation	RadAvg	1351	1597	1797	1899	1872	1973	1989	1929	1825	1593	1337	1281	
	RadStd	95	93	124	137	122	100	102	83	113	117	110	74	